Shape + colour + material on EVERY 3-D image (v0.81.1)

The parametric shape and the scan's colour/material/finish now apply consistently across all 3-D representations.

Shape (parametric)

- Family chosen from the part name — now broader:
- grommet→o-ring, filter/coupling/cable→tube,
- valve/plug/key→shaft, terminal/lug→bracket.
- Geometry generated from FLIS dimensions.

Actual shapes, not boxes.

Colour + material (scan)

- Full vocabulary in the client now: military colours,
- metals, polymers, platings/paint -> colour + finish.
- Mirrors the server material_feature exactly.
- Cards + SVG fallback + modal all read the same.

Looks like the real part.

Everywhere + correlated

- Applied to every collection card, the SVG fallback,
- and the modal (which then refines with authoritative
- FLIS material). Paired with the cited scan (Manual
- illustration) and the cross-reference record.

Consistent across the app.

The whole CAD experience, unified

Each 3-D item is now: a parametric shape (from the name) + dimensions (from FLIS) + colour/material/finish (from the scan's description) + its cited breakdown image + its cross-reference (vehicles/maker/interchange). Editable live in the Parametric CAD panel. Procedural and offline — not a scanned OEM solid (ITAR), but dimensionally + materially grounded and paired with the authoritative illustration.

VERIFIED

Client material port matches the server (rubber→matte black, steel+zinc→silvery metallic, olive-drab+paint→flat olive, brass→warm gloss); expanded classifier maps the new families; all 13 geometry families build valid meshes. Client-only ES5; WebGL with SVG fallback (RPS-safe).

BUILT diagram. Dark (R3). v0.81.1 · 2026-06-03 · threed.html appearance() full vocab · partgeo.js classifier · render3D material fallback.